



## 62 Instructor's Manual for Marine Biology

2. Variability in salinity stems from three central features: Salt water is more dense than freshwater; tides are periodic events; and the Coriolis effect causes a moving body to be deflected from a straight path.
3. Soft muds commonly dominate sediments in estuaries. Their composition and origin may be complex, involving both terrestrial and marine materials. Because coarser sediments settle out of the water column first, followed by fine-grained material, there is a gradient in sediment size that is dependent on current strength. In geologic parlance, this concept is called facies (for example, lithofacies).
4. Storms can play significant roles in redistributing sediments, reducing salinity, burying organisms, eroding the shoreline, etc.
5. Because of their smaller volume and the influence of river temperatures, estuaries have greater fluctuation in temperature than other inshore waters. They are usually protected by the shoreline from severe winds, so large waves rarely develop. The currents present in estuaries develop from the combination of tidal and river flows.
6. Estuaries are similar to most lakes in that they have inputs and outflows, the combination of which results in a flushing time that depends on the size of the basin, and the relative volume of inputs and outflows.

### III. The Biota of Estuaries

Marine, freshwater, and true brackish-water animals inhabit estuaries. This section is composed of three parts:

- Faunal Composition
- Estuarine Vegetation
- Estuarine Plankton

|                 |                     |                    |
|-----------------|---------------------|--------------------|
| brackish water  | salt marsh          | stenohaline        |
| estuarine water | sea grasses         | transitional fauna |
| euryhaline      | species composition |                    |

1. The marine component, of course, invades an estuary from the ocean side. This component is composed of two groups: stenohaline and euryhaline. The former tolerate little changes in salinity, while the latter can withstand greater variation. Few freshwater forms extend into estuaries from rivers and streams, and they are generally intolerant of salt, dropping out before the salinity reaches 10 psu. A true brackish-water or estuarine fauna is also depauperate when compared with the marine species present in these habitats. Some species are transitional, passing through the estuary as part of their life cycle.
2. Estuaries also are somewhat limited in their number of large plants. Sea grasses, macroalgae such as *Ulva*, and diatoms can be common. Various salt-marsh plants are present as well. (See Section VI.)
3. Diatoms are frequently the dominant members of the phytoplankton, but the composition and abundance of estuarine phytoplankton depend on factors such as turbulence, turbidity, and flushing rates. These same factors also are important in determining the composition of the zooplankton. Estuarine diatoms undergo bloom conditions, at which time they may shade sea grasses.

### IV. Adaptations of Estuarine Organisms

Organisms living in this stressful environment have evolved certain interesting morphological, physiological, and behavioral adaptations, as detailed in the three parts that compose this section:

- Morphological Adaptations
- Physiological Adaptations
- Behavioral Adaptations

|            |                       |                                  |
|------------|-----------------------|----------------------------------|
| aerenchyma | free amino acids      | intracellular osmotic regulation |
| anoxic     | hypertonic            | isotonic                         |
| cuticles   | hypotonic ion balance | lignin                           |

|                |                 |                     |
|----------------|-----------------|---------------------|
| migration      | osmosis         | succulence strategy |
| nursery        | salt glands     | water balance       |
| osmoconformers | <i>Spartina</i> |                     |
| osmoregulators | stomata         |                     |

1. Morphological adaptations specific to an estuarine environment are few. One such example is that some estuarine plants possess aerenchymal tissue that can bring oxygen down to roots buried in anoxic sediments.
2. The ability to tolerate fluctuations in salinity without significant tissue damage is one adaptation present in some estuarine animals. Such organisms are called osmoconformers. The osmotic concentration of their internal fluids fluctuates with the external environment. The converse of this physiology is that of the osmoregulator; this animal moves salts or otherwise adjusts ions to maintain a favorable water balance.
3. Vascular plants, of course, have a similar problem with salinity fluctuations that they solve in much the same way as osmoregulators: by inducing movement of salts. Glands that secrete salt, depositing salts on leaf surfaces, and shedding leaves loaded with salt are examples of how these plants cope with NaCl.
4. Certain behaviors also can help animals mediate the problem of fluctuating salinity. Some burrow to enter sediments where salinity fluctuations are less. Other species migrate to water of a specific salinity, depending on their ability to handle the amount of salt present.

### V. Ecology of Estuaries

Although algal primary productivity is generally low, estuaries have considerable amounts of organic matter. Much of the organic material is produced by salt-marsh plants and then carried into the estuary. Benthic algae that has been resuspended, either alive or dead, also can be important, as well as material brought to the estuary by rivers and streams. This section is composed of three units:

- Productivity, Organic Matter, and Food Sources
- Food Webs
- Plankton Cycles

|                       |                          |                    |
|-----------------------|--------------------------|--------------------|
| American estuary      | detritus                 | spring bloom       |
| carbon/energy (C/E)   | European estuary         | suspension feeders |
| competitive exclusion | fall bloom               | vacuuming          |
| deposit feeders       | interference competition | winter bloom       |
| detritivores          | primary production       |                    |

1. Estuaries are sinks for organic matter that was produced elsewhere and then transported to the estuary. Such material is termed allochthonous. In contrast, autochthonous material is produced on site and stays there. Salt-marsh plants are the most important sources of autochthonous organic matter, but benthic diatoms and algal mats also make significant contributions.
2. In European estuaries, most of the carbon/energy (C/E) is imported from the sea or river. As net importers of energy, European estuaries are allochthonously driven. In American estuaries, excess carbon/energy is produced and is exported to the sea. Because these estuaries are driven by their own C/E, they are referred to as autochthonously driven.
3. Detritus appears to be the basis of the estuarine food web, but the particles themselves are of minor importance as a source of C/E. More important are the bacteria and fungi, which colonize and decompose detrital particulates. Because estuaries have low primary productivity by phytoplankton, the food web is probably based on detritus and the microbes that colonize this material.
4. Suspension feeding, direct detritus feeding, and particle sorting are trades employed by a variety of estuarine worms, mollusks, and crustaceans. These organisms are in turn consumed by both invertebrate and vertebrate predators.

## VI. Salt Marshes

"Salt marshes are communities of emergent herbs, grasses, or low shrubs rooted in soils that are alternately inundated and drained by tidal action." This section of the chapter is divided into nine parts:

- Definitions and Characterization
- Environmental Characteristics
- Composition and Distribution
- Physiognomy, Development, and Zonation
- Causes of Zonation
- Productivity
- Consumer Control in Salt-Marsh Systems
- Trophic Interactions and Food Webs
- Human Impacts in Salt Marshes

|                             |                        |                  |
|-----------------------------|------------------------|------------------|
| aerenchymal tissue          | halophytes             | peat             |
| anaerobic                   | herbaceous angiosperms | physiognomy      |
| anoxic                      | high marsh             | salt marsh       |
| black rush zone             | ice                    | salt pan         |
| competition                 | low marsh              | salt tolerant    |
| competitive dominance       | marsh hay              | tidal creek      |
| cordgrasses                 | mutualism              | top-down control |
| dendritic drainage channels | patches                | zonation         |
| ecotone                     |                        |                  |

1. Halophytes dominate these intertidal regions, but unlike the estuarine environment proper, salt marshes are usually subject to more extreme variations in salinity and temperature.
2. Salt marshes are depauperate in species, probably due to the harsh environment (fluctuating salinity and temperature), but the species that inhabit these environments tend to enjoy a wide distribution. Seen from the air, salt marshes appear to be a sort of fractal, due to the dendritic pattern of the drainage channels and tidal creeks. From ground level, they appear to be a broad, flat, monotonous expanse of grasses (for example, *Juncus*, *Salicornia*, and *Spartina*).
3. Needless to say, salt marshes have zonation patterns that vary from place to place. The textbook examines these patterns. See also the references cited as follows and in the Enrichment Exercises:

Bertness, M., B.R. Silliman, and R. Jefferies. 2004. Salt marshes under siege: agricultural practices, land development and overharvesting of the seas explain complex ecological cascades that threaten our shorelines. *Am. Sci.* 92: 54–61.

Bertness, M.D. 1992. The ecology of a New England salt marsh. *Am. Sci.* 80: 260–268.

Bertness, M.D., and A.M. Ellison. 1987. Determinants of pattern in a New England salt marsh community. *Ecol. Mono.* 80: 260–268.

Costa, C.S.B., J.C. Marangoni, and A.M.G. Azevedo. 2003. Plant zonation in irregularly flooded salt marshes: relative importance of stress tolerance and biological interactions. *J. Eco.* 91: 951–965.

Ewanchuk, P.J., and M.D. Bertness. 2004. Structure and organization of a northern New England salt marsh plant community. *J. Eco.* 92: 72–85.

Howard, R.J., and I.A. Mendelssohn. 1999. Salinity as a constraint on growth of oligohaline marsh macrophytes: II. Salt pulses and recovery potential. *Am. J. Botany* 86: 795–806.

James, M.L., and J.B. Zedler. 2000. Dynamics of wetland and upland subshrubs at the salt marsh–coastal sage scrub ecotone. *Amer. Mid. Nat.* 143: 298–311.

Levine, J.M., J.S. Brewer, and M.D. Bertness. 1998. Nutrients, competition and plant zonation in a New England salt marsh. *J. Eco.* 86: 285–292.

McComb, A.J. 1995. *Eutrophic shallow estuaries and lagoons*. Boca Raton, FL: Lewis Publishers.

Padgett, D.E., and J.L. Brown. 1999. Effects of drainage and soil organic content on growth of *Spartina alterniflora* (Poaceae) in an artificial salt marsh mesocosm. *Am. J. Botany* 86: 697–702.

Shumway, S. 1991. *Secondary succession patterns in a New England salt marsh*. Dissertation, Brown University, Providence, RI.

Young, T., S. Komarow, and L. Deegan. 1999. Population size and summer home range of the green crab, *Carcinus maenas*, in salt creeks. *Biol. Bull.* 197: 297–299.

4. Primary productivity of salt marshes is quite high, with most of the production coming from the vascular plants. Estimates of productivity range from a low of 50 to a high of 3,000 g C/m<sup>2</sup>/yr. Algal production is lower, ranging from 20–60% of the vascular plants. However, the methodologies and assumptions used by some studies on the productivity of estuaries have been challenged as being inadequate for the task.
5. Salt marshes have a high productivity, and some of that C/E may be exported beyond the marsh. The amounts exported can be substantial: 10–50% of the offshore phytoplankton production. However, what is not understood is how much of an impact this material has on the secondary production of waters receiving the C/E. This material may sediment out rapidly or otherwise be unavailable to zooplankton. While, at one time, salt-marsh communities were thought to be detritus based, we now recognize that there is a strong component of consumer control. The text discusses the evidence for this shifting view.
6. In summary, salt marshes are ecotones between terrestrial and marine environments. They possess numerous microhabitats for plants and animals, but in general they have been treated as wastelands that should be reclaimed for “more fitting” use. Growing up on the south shore of Long Island, New York, in the mid-1960s, I remember watching miles and miles of productive salt marshes give way to housing developments. The first thing done in this abuse was to construct a dike. Then large pipes were put down to aid in draining the area, and following that, thousands of truckloads of fill and sand were added. The abuse was finished by the construction of housing developments with little architectural diversity. The textbook reports other abuses perpetrated on estuaries and salt marshes.
7. The concept of **ecosystem services** should be explored with your students. In terms of human needs, we can think of ecosystems as providing resources or services to various stakeholders. These resources might be as simple as seeds and animals in hunter–gatherer societies; they may be as complex as esthetics in developed societies; they may be as distasteful as places to dump trash, including general refuse or nuclear waste. Because several stakeholders often use the same ecosystem, conflicts often arise. For example, is a rain forest to be enjoyed as is, or is it to be cut down for wood? Moreover, some stakeholders cannot speak for themselves. That is, if we exploit an ecosystem for human needs, the other populations are unable to voice their opposition. The United Nation’s Millennium Ecosystem Assessment (MA) initiative addresses these concepts (Alcamo et al., 2003).

Alcamo, J., et al. 2003. *Ecosystems and human well-being: a framework for assessment*. Washington, DC: Island Press.

## ENRICHMENT EXERCISES

### Animal Behavior

Depending on your particular situation, you may wish to have your students investigate some aspects of estuarine-animal behavior. Some of the following ideas may be explored even in inland schools if a suitable marine aquarium is available (see also Chapter 6 for additional ideas):

1. Examine *Limulus polyphemus*’s locomotion, digging behavior, and righting behavior. Small (ca. 5–7 cm long to the base of the tail) specimens are available from various suppliers. If larger (adult size) specimens are available, check for the presence of *Bdelloura candida*, the “*Limulus* leech” (Platyhelminthes, Bdellouridae), on the undersurface. Are these organisms parasitic, phoretic, or commensal? Are they uniformly distributed on the animal? Are *Bdelloura* present on small *Limulus*? (See also Chapter 10.)

The following is an interesting paper regarding *Limulus* breeding:

Cohen, J.A., and H.J. Brockmann. 1983. Breeding activity and mate selection in the horseshoe crab, *Limulus polyphemus*. *Bull. Mar. Sci.* 33: 274–281.

2. Analyze burrowing behaviors. The polychaete worm (*Glycera* spp.), the bivalves *Ensis* and *Mercenaria*, and certain crabs offer students the opportunity to make observations of how the organisms burrows into sediments. Students can examine this process by using a variety of natural or artificial substrata.
3. The sea hare, *Aplysia brasiliana*, produces a water-soluble purple dye that it releases upon disturbance. Students can do a simple biochemical analysis of the stuff and determine whether it acts as a repellent to other marine animals.

Nolen, T.G., and P.M. Johnson. 2001. Defensive inking in *Aplysia* spp.: multiple episodes of ink secretion and the adaptive use of a limited chemical resource. *J. Exp. Bio.* 204: 1257–1268.

4. Examine *Uca*'s locomotion, digging behavior, sexual-display behavior, etc. If crabs, including *Uca*, are stimulated in some rather rough ways, they will autotomize a limb. A demonstration (for example, a video) of this process might be useful for starting a conversation on predation.

On the other hand, the opposite behavior—freeze behavior—also can be seen in this species. Male fiddlers, in particular, will go stiff when held by the palm of their large claw. (N.B.: These small crabs are still capable of inflicting a pinch that smarts!) Many crabs will do this; besides in *Uca*, I have seen this behavior in *Sesarma* (wharf crab), *Persephona* (purse crab), members of the family Pilumnidae (mud crabs), and even *Callinectes* (blue crab). Take special care with *Callinectes*: Approach from the rear, avoid those pinchers, and pick it up only by the swimming legs. You are duly warned!

Croll, G.A., and J.B. McClintock. 2000. An evaluation of lekking behavior in the fiddler crab *Uca* spp. *J. Exp. Mar. Bio. Eco.* 254: 109–121.

5. The following are some references on the physiology of marine animals:

Anon. 1981. *Marine invertebrates*. Washington, DC: National Academy Press.

Diehl, F.A., J.B. Feeley, and D.G. Gibson. 1971. *Experiments using marine animals*. Eastlake, OH: Aquarium Systems, Inc. [From Instant Ocean®; probably out of print.]

Hammen, C.S. 1980. *Marine invertebrates: comparative physiology*. Hanover, NH: University Press of New England.

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Adam, P. 1990. *Saltmarsh ecology*. Cambridge, UK: Cambridge University Press.

Bertness, M.D., P.J. Ewanchuk, and B.R. Silliman. 2002. Anthropogenic modification of New England salt marsh landscapes. *PNAS* 99: 1395–1398.

Brafield, A.E. 1978. *Life in sandy shores*. Studies in Biology No. 89. London: Edward Arnold.

Brown, A.C., and A. McLachlan. 1990. *Ecology of sandy shores*. New York: Elsevier.

Brunenmeister, S., N. Fotheringham, and S.B. Rothschild. 2001. *Gulf coast marine life: Texas, Louisiana, Mississippi, Alabama and Florida*. Houston: Lone Star Books.

Carefoot, T. 1977. *Pacific seashores: a guide to intertidal ecology*. Seattle: University of Washington Press.

Carson, R. 1955. *The Edge of the Sea*. New York: Houghton Mifflin Company.

Chapman, V.J. (ed.). 1977. *Ecosystems of the world: Volume 1—wet coastal ecosystems*. New York: Elsevier.

Coultas, C.L., and Y.P. Hsieh. (eds.). 1995. *Ecology and management of tidal marshes*. Delray Beach, FL: St. Lucie Press.

Eisma, D. 1998. *Intertidal deposits: river mouths, tidal flats, and coastal lagoons*. Boca Raton, FL: CRC Press.

Gosner, K.L., and R.T. Peterson. 1999. *A field guide to the Atlantic seashore*. New York: Houghton Mifflin Company.

Kennish, M.J. 1986. *Ecology of estuaries: Volume I—physical and chemical aspects*. Boca Raton, FL: CRC Press.

Kennish, M.J. 1986. *Ecology of estuaries. Volume II—biological aspects*. Boca Raton, FL: CRC Press.

Kramer, K.J.M. 1994. *Biomonitoring of coastal waters and estuaries*. Boca Raton, FL: CRC Press.

Mitsch, W.J., and J.G. Gosselink. 1993. *Wetlands*. 2nd ed. New York: John Wiley and Sons.

- Pomeroy, L.R., and R.G. Wiegert. (eds.). 1981. *The ecology of a salt marsh*. New York: Springer-Verlag.
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- Ricketts, E., and J. Calvin. 1985. *Between Pacific tides*. 5th ed. Revised by D.W. Phillips. Stanford, CA: Stanford University Press.
- Roy, P.S., et al. 2001. Structure and function of south-east Australian estuaries. *Est. Coast. Shelf Sci.* 53: 351–384.
- Ruppert, E.E., and R.S. Fox. 1988. *Seashore animals of the southeast*. Columbia, SC: University of South Carolina Press.
- Stilgoe, J.R. 1994. *Alongshore*. New Haven, CT: Yale University Press.
- Teal, J.M. 1991. *Life and death of the salt marsh*. New York: Ballantine Books.
- Thompson, J.D. 1991. The biology of an invasive plant. *BioScience* 41(6): 393–401.
- Turner, R.E., and Rabalais, N.N. 1991. Changes in Mississippi River water quality this century. *BioScience* 41(3): 140–147.
- Zedler, J.B., and Powell, A.N. 1993. Managing coastal wetlands. *Oceanus* 35(2): 19–28.

